How-To Library

Informatica

# Implementing Salesforce to Snowflake synchronization using Cloud Mass Ingestion Applications

© Copyright Informatica LLC 2021, 2026. Informatica, the Informatica logo, and Informatica Cloud Data Integration are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html

# Abstract

This article describes how to set up Mass Ingestion Applications and create your first application ingestion job to replicate data from a Salesforce source to a Snowflake target.

# Supported Versions

- Informatica Intelligent Cloud Services December 2021

# Table of Contents

Overview. . . . 2
Step 1: Set up Mass Ingestion Applications. . . . 3
Set up an organization. . . . 3
Download and install a Secure Agent. . . . 3
Configure the Database Ingestion service. . . . 6
Step 2: Create a project folder. . . . 8
Step 3: Download the connectors. . . . 8
Step 4: Configure the connections. . . . 9
Salesforce Mass Ingestion connection properties. . . . 9
Snowflake Cloud Data Warehouse V2 connection properties. . . . 12
Step 5: Configure an application ingestion task. . . . 13
Defining basic task information. . . . 14
Configuring the Salesforce source. . . . 15
Configuring the Snowflake target. . . . 18
Configuring schedule and runtime options. . . . 23
Step 6: Deploy the application ingestion task. . . . 25
Step 7: Run the application ingestion job. . . . 25

# Overview

The Software-as-a-Service (SaaS) and on-premise applications used in business organizations store large amounts of business-critical data on a daily basis. If you move the data stored in the applications to a system or repository that can handle large volumes of data, you can utilize it for various purposes, such as advanced data analytics and data warehousing.

Application Ingestion and Replication provides the ability to transfer large amounts of data from SaaS and on-premise applications to cloud-based data lakes and data warehouses. After the initial batch of data is transferred, Application Ingestion and Replication can replicate data changes continuously from the source applications to keep the data up to date in the targets.

This article explains the steps that are required to set up a Mass Ingestion Applications environment and then configure and implement an application ingestion task that replicates Salesforce data to a Snowflake target.

For more information about Mass Ingestion Applications, see Mass Ingestion Applications in the Cloud Mass Ingestion help.

# Step 1: Set up Mass Ingestion Applications

To set up Application Ingestion and Replication, perform the following tasks in Administrator:

1. Set up an organization on page 3.
2. Download and install a Secure Agent on page 3.
3. Configure the runtime environment on page 6.

## *Set up an organization*

An organization is a secure area within the Informatica Intelligent Cloud Services (IICS) repository that stores your licenses, user accounts, ingestion tasks, and information about jobs and security. You can set up an organization in Administrator.

To set up an organization, perform the following tasks:

1. On the **Organization** page, configure organization properties such as the organization name and address, authentication information, and notification email addresses.

   **Note:** Application Ingestion and Replication does not support sub-organizations.
2. On the **Licenses** page, verify that your organization has the appropriate licenses.
   For information about the licenses that are required to use Application Ingestion and Replication, contact Informatica Global Customer Support.
3. On the **Users**, **User Groups**, and **User Roles** pages, configure users, user groups, and user role permissions.

For more information about how to perform the tasks, see the help document for the page.

## *Download and install a Secure Agent*

A Secure Agent is a lightweight program that runs tasks and enables secure communication across the firewall between your organization and Informatica Intelligent Cloud Services.

Install the Secure Agent on all the systems from which you want to use Application Ingestion and Replication. You can install a Secure Agent on Windows or Linux.

When you download and install a Secure Agent, a Secure Agent group, also called a runtime environment, is created. If a Secure Agent group contains multiple Secure Agents, Application Ingestion and Replication uses the first Secure Agent that is available within the group.

**Note:** You cannot use Application Ingestion and Replication on a Hosted Agent or serverless runtime environment.

### Downloading and installing the Secure Agent on Windows

To install the Secure Agent on a Windows machine, download and run the Secure Agent installer and then register the agent.

Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine. If any other Secure Agent exists, you must uninstall it first.

**Tip:** To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Runtime Installer**.
3. Select **Secure Agent** as the Environment Type.
4. Select **Windows 64** as the Platform.

5. Click **Copy** to copy the install token.

   You will need this token to install the Secure Agent. The token expires after 24 hours. If the token expires before you install the Secure Agent, repeat steps 1 to 5 to obtain a new token. You don't need to download the installer again.

6. Click **Download**.

   The installer is downloaded to your machine. The name of the installer is `agent64_install_ng_ext.<agent core version>.exe`.

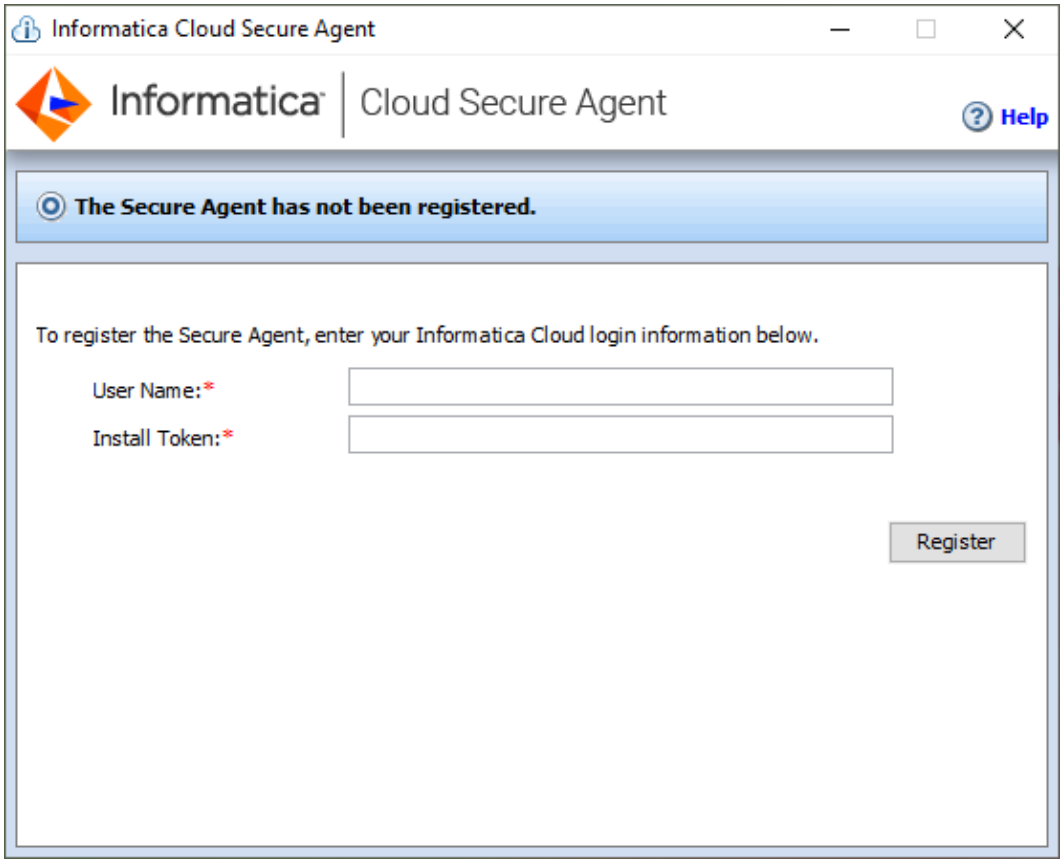
7. Run the installer as an Administrator and complete the following steps:

   a. Specify the Secure Agent installation directory, and click **Next**.

      **Note:** Ensure that the file path doesn't contain multibyte characters. If the path contains multibyte characters, the Secure Agent might not start.

   b. Click **Install** to install the agent.

   The **Cloud Secure Agent** dialog box opens and prompts you to register the agent as shown in the following image:



8. In the Secure Agent Manager, enter the following information, and then click **Register**:

| Option | Description |
|---|---|
| User Name | User name that you use to access Informatica Intelligent Cloud Services. |
| Install Token | Token that you copied when you downloaded the runtime installer. |

The Secure Agent Manager displays the status of the Secure Agent. It takes a minute for all of the services to start.

**Tip:** When you register the agent, it is added to its own Secure Agent group by default. You can add the agent to a different Secure Agent group if necessary.

9. If your organization uses an outgoing proxy server to connect to the internet, enter the proxy server information.
10. Close the Secure Agent Manager.

The Secure Agent Manager minimizes to the taskbar and continues to run as a service until stopped.

## Downloading and installing the Secure Agent on Linux

To install the Secure Agent on a Linux machine, download and run the Secure Agent installer and then register the agent.

Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine using the same Linux user account. If any other Secure Agent exists, you must uninstall it first.

**Tip:** To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Runtime Installer**.
3. Select **Secure Agent** as the Environment Type.
4. Select **Linux 64** as the Platform.
5. Click **Copy** to copy the install token.

You will need this token to install the Secure Agent. The token expires after 24 hours. If the token expires before you install the Secure Agent, repeat steps 1 to 5 to obtain a new token. You don't need to download the installer again.

6. Click **Download**.

The installer is downloaded to your machine. The name of the installer is `agent64_install_ng_ext.<agent core version>.bin`.

7. Save the installer to a directory on the machine where you want to run the Secure Agent.

**Note:** Ensure that the file path doesn't contain spaces or multibyte characters. If the file path contains spaces, the installation might fail. If the path contains multibyte characters, the Secure Agent might not start.

8. From a shell command line, navigate to the directory where you downloaded the installation program and enter the following command:

```
./agent64_install_ng_ext.bin -i console
```

9. When the installer completes, navigate to the following directory:

```
<Secure Agent installation directory>/apps/agentcore
```

10. To start the Secure Agent, enter the following command:

```
./infaagent.sh startup
```

The Secure Agent Manager starts. Register the agent using the user name that you use to access Informatica Intelligent Cloud Services and also supply the install token.

11. To register the agent, in the `<Secure Agent installation directory>/apps/agentcore` directory, enter one of the following commands using your Informatica Intelligent Cloud Services user name and the token that you copied:

- To add the agent to its own Secure Agent group, use the following command:

```
./consoleAgentManager.sh configureToken <user name> <install token>
```

- To add the agent to an existing Secure Agent group, use the following command:

```
./consoleAgentManager.sh configureTokenWithRuntime <user name> <install token>
<Secure Agent group name>
```

**Note:** If the command includes a Secure Agent group name that doesn't exist, the Secure Agent is not assigned to a group. Be sure to use a valid Secure Agent group name.

The following table lists the command options:

| Option | Description |
|---|---|
| User Name | Required. Informatica Intelligent Cloud Services user name of the user installing the Secure Agent. |
| Install Token | Required. The install token that you copied when you downloaded the runtime installer. |
| Secure Agent group name | Optional. Include a group name when you want to add the agent to an existing Secure Agent group instead. If this option isn't included in the command, the agent will be in its own Secure Agent group. |

You can check the registration status of a Secure Agent using the following command:

```
./consoleAgentManager.sh isConfigured
```

## *Configure the Database Ingestion service*

The Database Ingestion service is a microservice that is used by Secure Agent to run both application ingestion and replication and database ingestion and replication tasks. The Database Ingestion service is automatically downloaded to the system when you download the Secure Agent.

After you download the Secure Agent to your runtime environment and enable the Database Ingestion service, the Database Ingestion packages are pushed to the on-premises system where the Secure Agent runs. You can then optionally configure the properties and the environment variable for the Database Ingestion service that runs on the Secure Agent.

### Database Ingestion service properties for Mass Ingestion Applications

To optimize the behavior of the Database Ingestion service for Application Ingestion and Replication, configure the Database Ingestion service properties for your runtime environment.

To configure the properties, open your runtime environment and click **Edit**. Under **System Configuration Details**, select the **Database Ingestion** service and the **DBMI_AGENT_CONFIG** type.

The following table describes the Database Ingestion service properties that you can configure for Application Ingestion and Replication:

| Property | Description |
|---|---|
| maxTaskUnits | The maximum number of application ingestion and replication tasks and database ingestion and replication tasks that can run concurrently on the on-premises machine where the Secure Agent is running.<br>Each ingestion task requires at least 1 task unit to run.<br>Informatica recommends that you configure no more than 2.5 task units per core. For example, if you have an 8-core machine, you can set the maxTaskUnits property to 20. If you have a 16-core machine, you can set it to 40.<br>During initial load processing, this property determines the number of objects that can be unloaded simultaneously. Remaining objects are queued and then unloaded when resources become available.<br>During incremental load processing, this property determines the number of application ingestion and replication and database ingestion and replication jobs that can run simultaneously.<br>A single job can process many objects. The total number of objects that can be processed is limited only by available memory. On an average, 25 MB of RAM is required per object for an initial load task based on a 1 KB row size. |
| serviceLogRetentionPeriod | The number of days to retain each internal Database Ingestion service log file after the last update is written to the file. When the specified retention period elapses, the log file is deleted. The default value is 7 days.<br>The service logs are retained on the Secure Agent host where they are created: `<infaagent>/apps/Database_Ingestion/logs`. |
| taskLogRetentionPeriod | The number of days to retain each job log file after the last update is written to the file. When the specified retention period elapses, the log file is deleted. The default value is 7 days. |
| serviceUrl | The URL that the Database Ingestion service uses to connect to the Informatica Intelligent Cloud Services cloud. |
| logLevel | The level of detail to include in the logs that the Database Ingestion service creates. The options are:<br>- TRACE<br>- DEBUG<br>- INFO<br>- WARN<br>- ERROR<br>Default value is TRACE. |
| taskExecutionHeapSize | The maximum heap size, in gigabytes, for the Task Execution service. This value, in conjunction with maxTaskUnits property, affects the number of concurrent application ingestion and replication tasks that can run on a Secure Agent. You can increase the heap size to run more tasks concurrently. Enter this value followed by "g" for gigabytes, for example, '9g'. The default value is 8g. |
| useProxy | Set this property to true to enable the Database Ingestion service to go through a proxy when connecting to or writing data to targets. The service then uses the proxy settings from the Secure Agent proxy configuration. By default, proxy settings are not used. |
| testProperty | Do not set this property. It is intended for internal use by Informatica Global Customer Support and technical staff. This property appears only if you select DBMI_AGENT_ENV in the **Type** field. |

**Note:** The service properties described in the table are applicable to both Application Ingestion and Replication and Database Ingestion and Replication.

### Environment variables for Mass Ingestion Applications

To optimize the behavior of the Database Ingestion service for Application Ingestion and Replication, define the following environment variable:

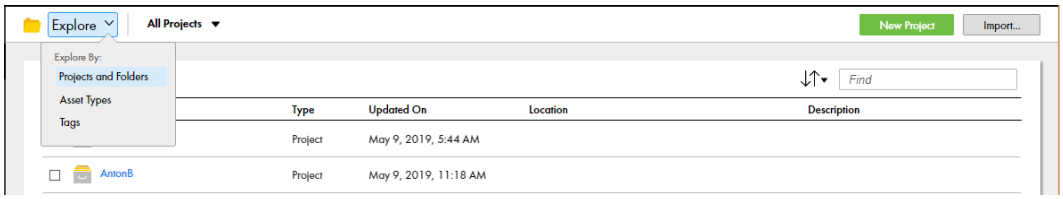
| Environment Variable | Description |
|---|---|
| DBMI_WRITER_CONN_POOL_SIZE | The number of connections that an application ingestion and replication job uses to propagate the change data to the target. The default value is 8. Valid values are 4 through 8. |

**Note:** After you define or change the environment variable, restart the Database Ingestion service for the changes to take effect.

## Step 2: Create a project folder

Create a folder in a project to logically organize your assets. When you define an application ingestion and replication task, you must specify the project or project folder location to store the task definition.

1. In the Data Ingestion and Replication, open the **Explore** page.
2. If the **Explore** page shows objects other than projects, select **Projects and Folders** in the **Explore** menu.




3. To create a project, click **New Project**.
4. In the **New Project Properties** dialog box, enter a project name up to 255 characters in length. You can also enter an optional description of the project.

   A project name cannot contain the following characters: `#?`|{}"^&[]/\.`
5. If you want to add a folder under the project, select the project and click **New Folder**.
6. In the **New Folder Properties** dialog box, enter a folder name up to 255 characters in length. You can also enter a brief description of the project.

   A folder name cannot contain the following characters: `#?`|{}"^&[]/\.`

   **Note:** You cannot create subfolders under a project folder.

## Step 3: Download the connectors

From the **Add-On Connectors** page in Administrator, download the following connectors if they are not already installed:

- Salesforce Mass Ingestion
- Snowflake Cloud Data Warehouse V2

# Step 4: Configure the connections

On the **Connections** page in Administrator, configure a connection for the Salesforce source and a connection for the Snowflake target.

1. In Administrator, select **Connections**.
2. On the **Connections** page, click **New Connection**.
3. Configure the following connection details:

| Property | Description |
|---|---|
| Connection Name | Name of the connection.<br>Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -<br>Maximum length is 100 characters. Connection names are not case sensitive. |
| Description | Brief description of the connection. |
| Type | Type of the connection, such as Salesforce Mass Ingestion. |

   After you select the connection type, additional properties that are specific to the connection type appear.

4. Configure the connection-specific properties.

   For descriptions of the connection properties, see Salesforce Mass Ingestion connection propertieson page 9 and Snowflake Cloud Data Warehouse V2 connection propertieson page 12.
5. To test the connection, click **Test Connection**.

   The results of the test appear at the top of the page.

   If a connection fails, contact the database administrator, or recheck your settings and verify that the selected runtime environment is up and running.
6. Click **Save**.

## *Salesforce Mass Ingestion connection properties*

When you set up a Salesforce Mass Ingestion connection, you must configure the connection properties.

Before you configure the connection, perform the following tasks:

- Create an External Client App in Salesforce to access data. For more information, see Create an External Client App.
- Configure the JWT Bearer OAuth flow for the External Client App. For more information, see Configure OAuth 2.0 JWT Bearer Flow for External Client Apps.
- Add the required OAuth scopes to the ECA to avoid connection test failures. At a minimum, include `Manage user data via APIs (api)` and `Perform requests at any time (refresh_token, offline_access)` scopes.

The properties of a Salesforce Mass Ingestion connection vary based on the authentication method that you specify for the connection. When you create a connection, you can select one of the following authentication methods:

- **OAuth 2.0 Username-Password Flow**: Authenticates the connection by using the Salesforce account login credentials and the consumer key and consumer secret that Salesforce generates for the External Client App.

  **Note:** Effective in the October 2026 release, OAuth 2.0 Username-Password Flow authentication in Salesforce Mass Ingestion Connector is deprecated because Salesforce is retiring the SOAP API login() call in API versions 31.0 through 64.0 to align with enhanced cybersecurity standards. For more information, see Platform SOAP API login() Retirement. Informatica intends to drop support for OAuth 2.0 Username-Password Flow on May 1, 2027. In the interim, transition to OAuth 2.0 JWT Bearer Flow authentication.

  **Deprecation Notice:** Deprecated functionality is supported, but Informatica intends to drop support in a future release. Informatica requests that you transition to different functionality before the functionality is dropped.

- **OAuth 2.0 JWT Bearer Flow**: Authenticates the connection by using the Salesforce account user name, private key alias, private key password, and the consumer key that Salesforce generates for the External Client App. Informatica recommends that you use this authentication method because this method provides secured access to Salesforce without sharing sensitive information, such as consumer secret and Salesforce account password.

## Connection properties for OAuth 2.0 Username-Password Flow authentication

The following table describes the connection properties for a Salesforce Mass Ingestion connection configured with OAuth 2.0 Username-Password Flow authentication:

| Connection property | Description |
|---|---|
| Connection Name | A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -<br>Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive. |
| Description | An optional description for the connection. Maximum length is 255 characters. |
| Runtime Environment | Name of the runtime environment where you want to run the ingestion tasks.<br>For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent. |
| User Name | User name of the Salesforce account. |
| Password | Password for the Salesforce account. |
| Security Token | Security token associated with the Salesforce account.<br>You can configure the connection without specifying the security token if there are no IP restrictions specified for the External Client App. However, you must specify the security token if IP restrictions are enforced for the app and if the Secure Agent is not running on the trusted IP range specified for your Salesforce organization.<br>**Note:** If you do not have the security token, reset the security token in Salesforce. For more information about resetting the security token, see the Salesforce documentation. |
| Consumer Key | Consumer key that Salesforce generates when you enable OAuth 2.0 authentication for the External Client App. |
| Consumer Secret | Consumer secret that Salesforce generates when you enable OAuth 2.0 authentication for the External Client App. |

| Connection property | Description |
|---|---|
| API Version | Version of the Salesforce API that you want to use to access the source data.<br>Default is 51.0.<br>**Note:** You cannot use a version older than 51.0. |
| OAuth token URL | OAuth 2.0 token endpoint of the Salesforce organization. The External Client App sends access token requests to this endpoint.<br>Default value is:<br>`https://login.salesforce.com/services/oauth2/token`<br>This default URL is used for all Salesforce instances.<br>Alternatively, you can enter an instance-specific URL:<br>`https://<instance domain URL>/services/oauth2/token`<br>An instance-specific URL can establish a more direct and faster connection to the Salesforce host server. If the load on the common default endpoint is heavy and ingestion jobs fail with an authentication error when using it, use this alternative URL instead. |

**Note:** For more information about the OAuth 2.0 Username-Password Flow authentication method, see the Salesforce documentation.

### Connection properties for OAuth 2.0 JWT Bearer Flow authentication

The following table describes the connection properties for a Salesforce Mass Ingestion connection configured with OAuth 2.0 JWT Bearer Flow authentication:

| Connection property | Description |
|---|---|
| Connection Name | A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -<br>Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive. |
| Description | An optional description for the connection. Maximum length is 255 characters. |
| Runtime Environment | Name of the runtime environment where you want to run the ingestion tasks.<br>For application ingestion and replication tasks, you can select a Secure Agent or a serverless runtime environment. You can't use a Hosted Agent. |
| User Name | User name of the Salesforce account. |
| Consumer Key | Consumer key that Salesforce generates when you enable OAuth 2.0 authentication for the External Client App. |
| Keystore Path | Absolute path to the keystore file that contains the X509 certificate required to validate a JSON Web Token (JWT) and establish a secure connection with Salesforce.<br>The keystore file must be in the Java KeyStore (JKS) format. |
| Keystore Password | Password for the keystore file. |
| Private Key Alias | Alias name of the private key used to sign the JWT. |

| Connection property | Description |
|---|---|
| Private Key Password | Password for the private key. |
| API Version | Version of the Salesforce API that you want to use to access the source data.<br>Default is 51.0.<br>**Note:** You cannot use a version older than 51.0. |
| OAuth token URL | OAuth 2.0 token endpoint of the Salesforce organization. The External Client App sends access token requests to this endpoint.<br>Default value is:<br>`https://login.salesforce.com/services/oauth2/token`<br>This default URL is used for all Salesforce instances.<br>Alternatively, you can enter an instance-specific URL:<br>`https://<instance domain URL>/services/oauth2/token`<br>An instance-specific URL can establish a more direct and faster connection to the Salesforce host server. If the load on the common default endpoint is heavy and ingestion jobs fail with an authentication error when using it, use this alternative URL instead.<br>If you plan to use Bulk API 1.0 in your task to ingest data from a Salesforce source, you need to specify the instance-specific URL. |

**Note:** For more information about the OAuth 2.0 JWT Bearer Flow authentication method, see the Salesforce documentation.

## *Snowflake Cloud Data Warehouse V2 connection properties*

When you set up a Snowflake Cloud Data Warehouse V2 connection, you must configure the connection properties.

When you create a connection, select the Standard authentication method.
**Note:** Mass Ingestion Applications does not support the Authorization Code and KeyPair authentication methods.

### Connection properties for Standard authentication

The following table describes the connection properties for a Snowflake Cloud Data Warehouse V2 connection configured with Standard authentication:

| Connection property | Description |
|---|---|
| Runtime Environment | The name of the runtime environment where you want to run the tasks.<br>Specify a Secure Agent as the runtime environment. |
| Authentication | Select the authentication method that the connector must use to log in to Snowflake.<br>Select **Standard**. Default is **Standard**. |
| Username | The user name to connect to the Snowflake Cloud Data Warehouse account. |
| Password | The password to connect to the Snowflake Cloud Data Warehouse account. |

| Connection property | Description |
|---|---|
| Account | The name of the Snowflake account.<br>In the Snowflake URL, your account name is the first segment in the domain.<br>For example, `123abc` is your account name in `https://123abc.snowflakecomputing.com`. |
| Warehouse | The Snowflake warehouse name. |
| Role | The Snowflake role assigned to the user. |
| Additional JDBC URL Parameters | The additional JDBC connection parameters.<br>Enter the following value in this field:<br>`database=<target_database_name>`<br>If you want to specify additional JDBC connection parameters, enter the values in the following format:<br>`<param1>=<value>&<param2>=<value>&<param3>=<value>....`<br>For example:<br>`user=jon&warehouse=mywh&db=mydb&schema=public`<br>**Important:** Ensure that there is no space before and after = when you add the parameters.<br>Refer to the following examples of additional JDBC connection parameters that you can configure:<br>- To override the database and schema name used to create temporary tables in Snowflake, enter the database and schema name in the following format:<br>`ProcessConnDB=<DB name>&ProcessConnSchema=<schema_name>`<br>- To view only the specified database and schema while importing a Snowflake table, specify the database and schema name in the following format:<br>`database=<database_name>&schema=<schema_name>`<br>- To access Snowflake through Okta SSO authentication, enter the web-based IdP implementing SAML 2.0 protocol in the following format:<br>`authenticator=https://<Your_Okta_Account_Name>.okta.com`<br>**Note:** Microsoft ADFS is not applicable.<br>For more information about configuring Okta authentication, see the Snowflake website. |

# Step 5: Configure an application ingestion task

Use the application ingestion and replication task wizard in Mass Ingestion to configure application ingestion and replication tasks.

To configure an application ingestion task, perform the following tasks on the wizard:

1. Define basic task information on page 14.
2. Configure the Salesforce source on page 15.
3. Configure the Snowflake target on page 18.
4. Configure the task schedule and runtime options on page 23.

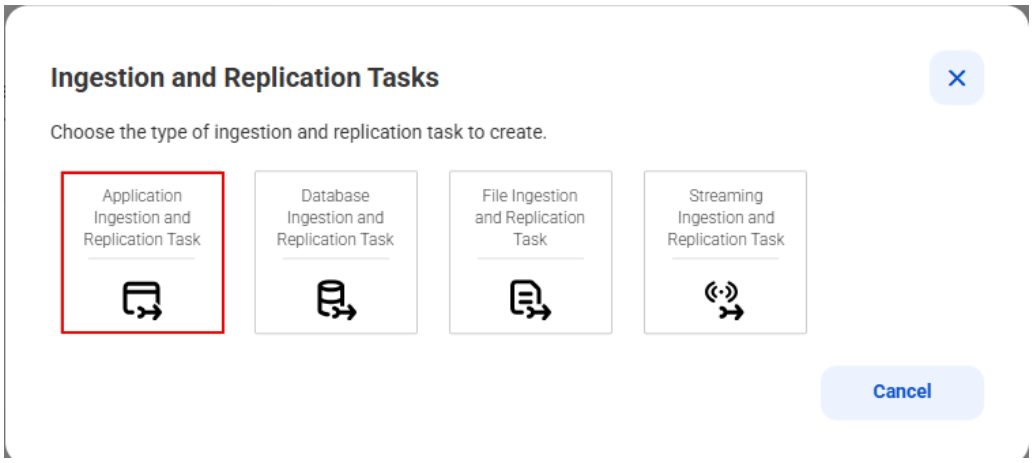
Click **Next** or **Back** to navigate from one page to another. At any point, you can click **Save** to save the information that you have entered until then.

After you complete all the wizard pages, save the information and then click **Deploy** to make the task available as an executable job to the Secure Agent.

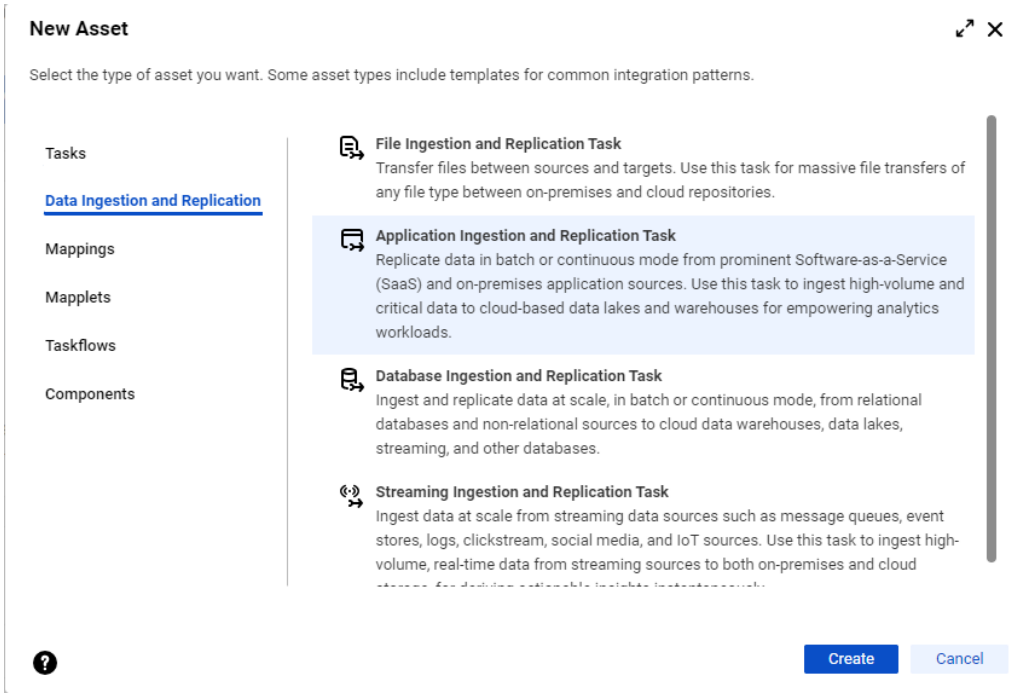
## *Defining basic task information*

To define an application ingestion and replication task, you must first enter some basic information about the task, such as the task name, project or project folder location, and load operation type.

1. Start the task wizard in one of the following ways:
   - On the Home page, click the **Ingest** panel and select **Application Ingestion and Replication Task**.




   - In the navigation bar on the **Explore** page or the Home page, click **New** to open the **New Asset** dialog box. Then, select **Data Ingestion and Replication** > **Application Ingestion and Replication** and click **Create**.




     **Note:** If your organization does not have a custom license for application, database, file, or streaming ingestion and replication, the **Data Ingestion and Replication** category still appears but you cannot select and configure an ingestion and replication task of the unlicensed task type or types.

   The **Definition** page of the application ingestion and replication task wizard appears.

2. Configure the following properties:

| Property | Description |
|---|---|
| Name | Name of the application ingestion and replication task.<br>The name of the application ingestion and replication task must be unique within the organization. The name can contain alphanumeric characters, spaces, periods (.), commas (,), underscores (_), plus signs (+), and hyphens (-).<br>Task names are not case sensitive. The maximum length is 50 characters.<br>**Note:** If you include spaces in the name of an application ingestion and replication task, the spaces do not appear in the name of the job associated with the task. |
| Location | Project or folder in which you want to store the task. |
| Runtime Environment | The runtime environment in which you want to run the task.<br>The runtime environment must be a Secure Agent group that consists of one or more Secure Agents. A Secure Agent is a lightweight program that runs tasks and enables secure communication.<br>For application ingestion and replication tasks, the Cloud Hosted Agent is not supported and does not appear in the **Runtime Environment** list.<br>You can use serverless runtime environments hosted on Microsoft Azure for application ingestion and replication initial load jobs only for some sources.<br>**Tip:** Click the **Refresh** icon to refresh the list of runtime environments. |
| Description | A brief description of the task.<br>Maximum length is 4000 characters. |
| Load Type | Type of load operation that you want the application ingestion and replication task to perform. You can select one of the following load types for the task:<br>- **Initial Load**: Loads data read at a specific point in time from the source application to the target in a batch operation. You can perform an initial load to materialize a target to which incremental change data will be sent.<br>- **Incremental Load**: Propagates source data changes to a target continuously or until the job is stopped or ends. The job propagates the changes that have occurred since the last time the job ran or from a specific start point for the first job run.<br>- **Initial and Incremental Load**: Performs an initial load of point-in-time data to the target and then automatically switches to propagating incremental data changes made to the same source objects on a continuous basis. |

3. Click **Next**.

## *Configuring the Salesforce source*

On the **Source** page of the application ingestion and replication task wizard, configure the source.

1. From the **Connection** list, select the connection that you configured for the Salesforce source.

   **Note:** After you deploy the ingestion task, you cannot change the connection without undeploying the associated ingestion job. After you change the connection, you must deploy the task again.

2. In the **Object Selection Rules** section, create object selection rules to select the source objects that you want to load to the target.

   By default, an *Include* rule configured to select all objects on the source is defined in the task. If you do not want to load all the source objects to the target, you can define *Include* rules and *Exclude* rules to select the specific objects that you want to load.

Perform the following steps to create an object selection rule:

a. Select **Object Selection** as the rule type.

b. From the adjacent list, select **Include** or **Exclude** as the action that you want the rule to perform.

c. In the condition field, enter an object name or an object-name mask to specify the source objects to include in or exclude from the list of selected objects.

**Notes:**

- A mask can contain the asterisk (*) wildcard character to represent one or more characters, the question mark (?) wildcard character to represent a single character, or both types of wildcard characters. You can use a wildcard character multiple times in an object-name mask.
- Object names are case sensitive. When you define the object selection rules, you must specify the object names or masks in the case in which they are defined on the source.
- If an object name contains delimiters, such as quotation marks or brackets, do not include them when you specify the object name for the rule.
- If an object name includes special characters, such as backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), replace each special character in the name with a backslash (\) when you specify the object name for the rule.

d. Click **Add Rule**.

The object selection rules are processed in the order in which they are listed in the **Object Rules** list. The rule at the top of the list is processed first. You can use the arrow icons to change the order in which the rules are listed. For an example of using multiple rules, see Examples of rules selecting source objectson page 18.

After you create the rules, you can optionally click **Object Count** to display the number of source objects that match each rule in the **Object Affected** column and the total number of objects selected based on all the selection rules in the **Total Objects** field.

3. In the **Object Selection Rules** section, create data field action rules to perform actions on the data fields of the selected objects.

Perform the following steps to create a data field action rule:

a. Select **Data Field Action** as the rule type.

b. From the adjacent list, select one of the following action types:

- **LTRIM**. Trims spaces to the left of character data field values.
- **RTRIM**. Trims spaces to the right of character data field values.
- **TRIM**. Trims spaces to the left of and to the right of character data field values.

c. In the condition field, enter a data field name or a data field-name mask. The value that you enter is matched against data fields of the selected source objects to identify the data fields to which the action applies.

d. Click **Add Rule**.

**Note:** The data field action rules are processed in the order in which they are listed in the **Data Field Actions** list. The rule at the top of the list is processed first. You can use the arrow icons to change the order in which the rules are listed.

4. Perform the following steps to download a list of source objects that match the selection rules:

a. From the **List Objects** list, select the type of selection rule for which you want to download the list of source objects.

b. If you want to include the fields in the list, select **Include Fields**.

c. Click the Download icon.

The list of source objects that match the selection rules is downloaded to your local drive. The information in the downloaded list is in the following format:

*status*,*object_name*,*object_type*,*field_name*,*comment*

The following table describes the information in the downloaded file:

| Field | Description |
|---|---|
| status | Indicates whether Application Ingestion and Replication includes or excludes the source object from processing. The possible values are:<br>- **E**. The object is excluded from processing by an *Exclude* rule.<br>- **I**. The object is included for processing.<br>- **X**. The object is excluded from processing because its data type is not supported by Application Ingestion and Replication. |
| object_name | Name of the source object. |
| object_type | Type of the source object. The possible values are:<br>- **O**: Object.<br>- **F**: Field. |
| field_name | Name of the source field. This information appears only if you selected **Include Fields** before downloading the list. |
| comment | Reason why a source object is excluded from processing even though it matches the selection rules. |

5. In the **Fetch Size** field of the **Advanced** section, enter the number of records that the application ingestion and replication job associated with the task must read at a time from the source. The default value for initial load operations is 50000 and the default value for incremental load operations is 2000.
6. For incremental load tasks and combined initial and incremental load tasks, perform the following steps in the **Advanced** section:
   a. In the **Restart Point for Incremental Load** field, specify the point in the source data stream from which the ingestion job associated with the application ingestion and replication task starts extracting source change records.

      For incremental load tasks Salesforce sources, you must specify a date and time as the restart point. The ingestion job retrieves the changes that were made after the specified date and time, and then loads them to the target. You must specify the time in Greenwich Mean Time (GMT).

      For combined initial and incremental load tasks, Application Ingestion and Replication retrieves the change records from the latest available position in the data stream.

   b. In the **CDC Interval** field, specify the time interval in which the application ingestion and replication job must run to retrieve the change records for incremental load. Default value is 5 minutes.
7. In the **Custom Properties** section, you can specify custom properties that Informatica provides for special cases. To add a property, click the **Add Property** icon, and then add the property name and value.

   The custom properties are configured to address unique environments and special use cases.

   **Note:** Specify the custom properties only at the direction of Informatica Global Customer Support.
8. Click **Next**.

## Example of rules for selecting source objects

When you define a source for an application ingestion and replication task, you can define object selection rules to select the source objects that you want to load to the target. The following example demonstrates how you can use selection rules to select the required objects.

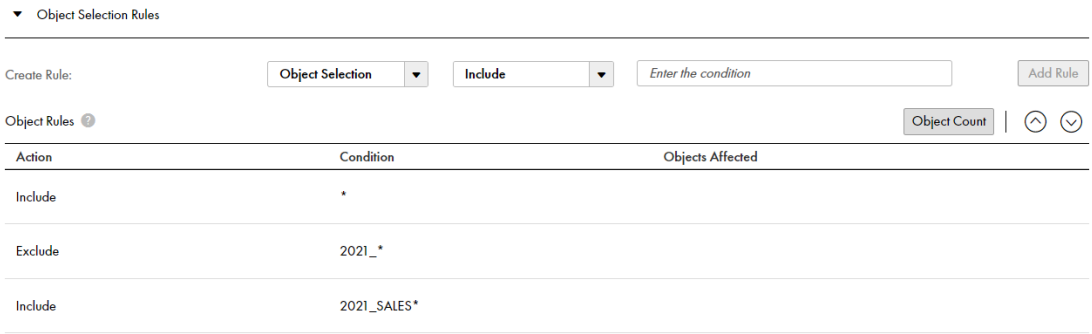
**Example**

A source has 1,000 objects with different prefixes. You want to select the objects that have the prefix "2021_SALES" and all objects with other prefixes except "2021_".

Define the following rules in the order in which they are listed:

- Rule 1: **Rule Action=Include** and **Condition=*** includes all objects on the source. When only the asterisk (*) wildcard is specified, all objects on the source are selected.
- Rule 2: **Rule Action=Exclude** and **Condition=2021_*** excludes the source objects that have names with the prefix "2021_".
- Rule 3: **Rule Action=Include** and **Condition=2021_SALES*** includes the source objects that have names with the prefix "2021_SALES".

The following image shows the rules in the **Object Selection Rules** section of the **Source** page:




# *Configuring the Snowflake target*

On the **Target** page of the application ingestion and replication task wizard, configure the target.

1. From the **Connection** list, select the connection that you configured for the Snowflake target.

   **Note:** After you deploy the ingestion task, you cannot change the connection without undeploying the associated ingestion job. After you change the connection, you must deploy the task again.
2. Configure the Snowflake properties.

   For descriptions of the properties, see Snowflake target propertieson page 19.
3. If you want to rename the target objects that are associated with the selected source objects, define table renaming rules.

   For more information about the table renaming rules, see “Renaming tables on the target” on page 22.
4. If you want to override the default mappings of source data types to target data types, perform the following steps in the **Data Type Rules** section to define data type rules:
   a. In the **Create Rule** fields, enter the source data type for which you want to customize the mapping and then enter the target data type that you want to map to the source data type.

**Important:**

- Application Ingestion and Replication does not support the BYTE and CHAR semantics in data-type mappings rules.
- If a source data type has a default value, you must specify it in your rule.

b. Click **Add Rule**.

The rule is created and appears in the rules list.

**Note:**

- You can define multiple data type mapping rules for a target. However, you can define only one rule for a source data type.
- After you deploy a task with custom mapping rules, you cannot edit the rules without undeploying the task.

5. In the **Custom Properties** section, you can specify custom properties that Informatica provides for special cases. To add a property, click the **Add Property** icon, and then add the property name and value.

   The custom properties are configured to address unique environments and special use cases.

   **Note:** Specify the custom properties only at the direction of Informatica Global Customer Support.

6. Click **Next**.

## Snowflake Data Cloud target properties

When you define an application ingestion and replication task, you must specify the properties for your Snowflake target on the **Target** page of the task wizard.

The following table describes the Snowflake target properties that appear in **Target** section:

| Property | Description |
|---|---|
| Target Creation | The only available option is **Create Target Tables**, which generates the target tables based on the source objects. |
| Schema | Select the target schema in which Application Ingestion and Replication creates the target tables. |

| Property | Description |
|---|---|
| Stage | The name of internal staging area that holds the data read from the source before the data is written to the target tables. This name must not include spaces. If the staging area does not exist, it will be automatically created.<br>**Note:** This field is not available if you selected the **Superpipe** option in the **Advanced Target Properties**. |
| Apply Mode | For incremental load and combined initial and incremental load jobs, indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. Options are:<br>- **Standard**. Accumulate the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target.<br>- **Soft Deletes**. Apply source delete operations to the target as soft deletes. A soft delete marks the deleted row as deleted without actually removing it from the database. For example, a delete on the source results in a change record on the target with "D" displayed in the INFA_OPERATION_TYPE column.<br>After enabling Soft Deletes, any update in the source table during normal or backlog mode results in the deletion of the matching record, insertion of the updated record, and marking of the INFA_OPERATION_TYPE operation as NULL in the target table. Similarly, inserting a record in the source table during backlog mode results in marking the INFA_OPERATION_TYPE operation as E in the target table record.<br>Consider using soft deletes if you have a long-running business process that needs the soft-deleted data to finish processing, to restore data after an accidental delete operation, or to track deleted values for audit purposes.<br>- **Audit**. Apply an audit trail of every DML operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select under the **Advanced** section. The audit columns contain metadata about the change, such as the DML operation type, transaction ID, and before image. Consider using **Audit** apply mode when you want to use the audit history to perform downstream computations or processing on the data before writing it to the target database or when you want to examine metadata about the captured changes.<br>After enabling the Audit apply mode, any update in the source table during backlog or normal mode results in marking the INFA_OPERATION_TYPE operation as E in the target table record. Similarly, inserting a record in the source table during backlog mode results in marking the INFA_OPERATION_TYPE operation as E in the target table record.<br>**Note:** The **Audit** apply mode applies for SAP source with SAP Mass Ingestion connector.<br>Default is **Standard**. |

Under **Advanced**, you can enter the following advanced target properties:

| Field | Description |
|---|---|
| Add Last Replicated Time | Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp, except for Snowflake targets that use the Superpipe option where minutes and seconds might vary slightly. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target.<br>By default, this check box is not selected. |
| Add Operation Type | Add a metadata column that includes the source SQL operation type in the output that the job propagates to the target tables. The column is named INFA_OPERATION_TYPE by default.<br>This field is displayed only when the **Apply Mode** option is set to **Audit** or **Soft Deletes**.<br>In Audit mode, the job writes "I" for inserts, "U" for updates, "E" for upserts, or "D" for deletes to this metadata column.<br>In Soft Deletes mode, the job writes "D" for deletes or NULL for inserts and updates. When the operation type is NULL, the other "Add Operation..." metadata columns are also NULL. Only when the operation type is "D" will the other metadata columns contain non-null values.<br>By default, this check box is selected. You cannot deselect it. |
| Add Operation Time | Select this check box to add a metadata column that records the source SQL operation timestamp in the output that the job propagates to the target tables.<br>This field is available only when **Apply Mode** is set to **Audit** or **Soft Deletes**.<br>By default, this check box is not selected. |
| Add Operation Sequence | Select this check box to add a metadata column that records a generated, ascending sequence number for each change operation that the job inserts into the target tables. The sequence number reflects the change stream position of the operation.<br>This field is available only when **Apply Mode** is set to **Audit**.<br>By default, this check box is not selected. |
| Add Before Images | Select this check box to add _OLD columns with UNDO "before image" data in the output that the job inserts into the target tables. You can then compare the old and current values for each data column. For a delete operation, the current value will be null.<br>This field is available only when **Apply Mode** is set to **Audit**.<br>By default, this check box is not selected. |
| Add Cycle ID | Select this check box to add a metadata column that includes the cycle ID of each CDC cycle in each target table. A cycle ID is a number that's generated by the CDC engine for each successful CDC cycle. If you integrate the job with Data Integration taskflows, the job can pass the minimum and maximum cycle IDs in output fields to the taskflow so that the taskflow can determine the range of cycles that contain new CDC data. This capability is useful if data from multiple cycles accumulates before the previous taskflow run completes. By default, this check box is not selected.<br>**Note:** If you select this option, you can't also select the Superpipe option for the Snowflake target. |
| Prefix for Metadata Columns | Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns.<br>The default value is INFA_. |

| Field | Description |
|---|---|
| Superpipe | Select this check box to use the Snowpipe Streaming API to quickly stream rows of data directly to Snowflake Data Cloud target tables with low latency instead of first writing the data to stage files. This option is available for all load types.<br>When you configure the target connection, select KeyPair authentication.<br>By default, this check box is selected. Deselect it if you want to write data to intermediate stage files.<br>**Note:** If you enable the **Superpipe** option for a task that uses the **Soft Deletes** apply mode, make sure the source tables contain a primary key. |
| Merge Frequency | When **Superpipe** is selected, you can optionally set the frequency, in seconds, at which change data rows are merged and applied to the Snowflake target tables.<br>The merge frequency affects how often the stream change data is merged to the Snowflake base table. A Snowflake view joins the stream change data with the base table. Set this value to balance the costs of merging data to the base table with the performance of view join processing.<br>This field applies to incremental load and combined initial and incremental load tasks. Valid values are 60 through 604800 seconds. Default is 3600 seconds. |
| Enable Case Transformation | By default, target table names and column names are generated in the same case as the corresponding source names, unless cluster-level or session-level properties on the target override this case-sensitive behavior. If you want to control the case of letters in the target names, select this check box. Then select a **Case Transformation Strategy** option. |
| Case Transformation Strategy | If you selected **Enable Case Transformation**, select one of the following options to specify how to handle the case of letters in generated target table (or object) names and column (or field) names:<br>- **Same as source**. Use the same case as the source table (or object) names and column (or field) names.<br>- **UPPERCASE**. Use all uppercase.<br>- **lowercase**. Use all lowercase.<br>The default value is **Same as source**.<br>**Note:** The selected strategy will override any cluster-level or session-level properties on the target for controlling case. |

## Renaming tables on the target

When you configure a target with an existing schema, you can optionally define rules for renaming the target tables that correspond to the selected source objects.

To create a rule for renaming tables, perform the following steps in the **Table Renaming Rules** section:

1. In the **Create Rule** fields, enter the name of the source object that you want to rename and then enter the name that you want to assign to the target table corresponding to the object.

   **Notes:**

   - You can enter only the asterisk (*) wildcard character to select all source objects that match the selection criteria defined on the **Source** page. Alternatively, you can enter the name of a specific source object or an object-name pattern that includes the asterisk (*) wildcard character.
   - If an object or table name includes special characters such as a backslash (\), asterisk(*), dot (.), or question mark (?), replace each special character in the name with a backslash (\) when you create the rule.
   - If you want to use an table-name pattern with a wildcard character for the target table, you must also use the wildcard character in the name of the corresponding source object.

- On Windows, if you enter target table renaming criteria that causes a target table name to exceed 232 characters in length, the name is truncated to 222 characters. Application Ingestion and Replication appends 14 characters to the name to add a date-time yyyyMMddHHmmss value, which causes the name to exceed the Windows maximum limit of 255. Ensure that the names of any renamed target tables will not exceed 232 characters.

2. Click **Add Rule**.
   The rule is created and appears in the rules list.

You can define multiple table renaming rules. Unless a table matches multiple rules, the order in which the rules are processed does not depend on the order in which they are listed in the **Table Renaming Rules** section. If a table matches multiple rules, the last matching rule determines the name of the table.

To delete a rule, click the Delete icon on the row that contains the rule.

**Example**

You want to add the prefix "PROD_" to the names of target tables that are associated with all selected source objects. In the **Table Renaming Rules** section, enter the following values in the **Create Rule** fields:

- For the source, enter the asterisk (*) wildcard character to specify all the source objects that match the object selection rules defined on the **Source** page.
- For the target, enter PROD_* to add this prefix to the names of all target tables corresponding to the source objects.

## *Configuring schedule and runtime options*

On the **Schedule and Runtime Options** page in the application ingestion and replication task wizard, you can specify a schedule for running the initial load jobs and configure the runtime options for jobs of all load types.

1. In the **Schema Drift Options** section, specify the schema drift option to use for each type of Data Definition Language (DDL) operation.

   **Note:** The **Schema Drift Options** section appears only for incremental load and combined initial and incremental load tasks.

   Application Ingestion and Replication supports the following types of DDL operations:

   - Add Field
   - Modify Field
   - Drop Field
   - Rename Field

The following table describes the schema drift options that you can specify for the DDL operations:

| Option | Description |
|---|---|
| Ignore | Does not replicate DDL changes that occur on the source schema to the target. For Snowflake targets, this option is the default option for the Drop Field and Rename Field operation types. |
| Replicate | Allows the application ingestion and replication job to replicate the DDL changes to the target. For Snowflake targets, this option is the default option for the Add Field and Modify Field operation types.<br>**Note:**<br>- If you try to replicate an unsupported schema change type on the target, the application ingestion and replication jobs associated with the task will fail with an error.<br>- Add Field operations that add a primary-key field are not supported and might cause unpredictable results.<br>- Modify Field operations that change the NULL or NOT NULL constraint of a field are not replicated to the target. |
| Stop Job | Stops the application ingestion and replication job. |
| Stop Object | Stops processing the source object on which the DDL change occurred.<br>**Note:** When one or more objects are excluded from replication because of the Stop Object schema drift option, the status of the job changes to **Running with Warning**. The application ingestion and replication job cannot retrieve the data changes that occurred on the source object after the job stops processing the changes. This action leads to data loss on the target. To avoid data loss, you must re-synchronize the source and target objects that the job stopped processing before you resume the application ingestion and replication job. |

2. Optionally, in the **Advanced** section, modify the value in the **Number of Rows in Output File** value to specify the maximum number of rows that the application ingestion and replication task writes to an output file on the target.

   Valid values are 1 through 100000000 and the default value is 100000 rows.

   **Note:** For incremental load and combined initial and incremental load operations, change data is flushed to the target either when the specified number of rows is reached or when the flush latency period expires and the job is not in the middle of processing a transaction. The flush latency period is the time that the job waits for more change data before flushing data to the target. The latency period is set to 10 seconds and cannot be changed.

3. If you want the application ingestion and replication job associated with the task to run in specific intervals based on a schedule, select **Run this task based on a schedule** in the **Schedule** section, and then select a predefined schedule for the job.

   By default, **Do not run this task based on a schedule** is selected, which configures the job to run only when it is manually triggered.

   **Note:** This field is available only for initial load tasks.

   You can view and edit the job schedule options in Administrator. If you edit the schedule, the changes are automatically applied to all the jobs that are configured to run based on the schedule. If you change the schedule for a task that is already deployed, the updated schedule is automatically applied to the application ingestion and replication job associated with the task.

   If a job is about to be triggered based on its schedule when its previous run is still in progress, Application Ingestion and Replication does not run the job and allows the job run that is already in progress to complete.

4. In the **Custom Properties** section, you can specify custom properties that Informatica provides for special cases. To add a property, click the **Add Property** icon, and then add the property name and value.

   The custom properties are configured to address unique environments and special use cases.

   **Note:** Specify the custom properties only at the direction of Informatica Global Customer Support.

## Step 6: Deploy the application ingestion task

After you define the application ingestion and replication task, deploy the task to create an executable job instance on the on-premises system that contains the Secure Agent and the Database Ingestion agent service. You can run the application ingestion and replication job only after you deploy the associated task. When you deploy the task, Application Ingestion and Replication also validates the task definition.

If you undeploy the job and then want to run the job again, you must deploy the task again to create a new job instance. The new job instance name ends with an incremented number in the format *taskname-job_instance_number*. The job instance number is incremented each time you deploy the ingestion task by adding 1 to the maximum instance number across all ingestion jobs.

Before you deploy the task, drop the existing target tables that do not match the structure of source objects. The existing target tables might not match the structure of source objects because of newly added source fields, dropped source or target fields, or modified field null constraints or data types. When you deploy the task after dropping the existing target tables, a new set of target tables are generated based on the source object selection rules and target table renaming rules.

▶ To deploy a task, in the application ingestion and replication task wizard, save the task and then click **Deploy**.

  If you included spaces in the name of the application ingestion and replication task, the spaces are omitted from the name of the corresponding application ingestion and replication job.

After you successfully deploy a task, an application ingestion and replication job is created and the status of the job is Deployed. You can run the job from the **My Jobs** page in Mass Ingestion or from the **All Jobs** tab on the **Mass Ingestion** page in Monitor.

If the deployment fails, the status of the corresponding application ingestion and replication job is Failed. To diagnose the error, you can download the error log from the **My Jobs** page in Mass Ingestion or from the **All Jobs** tab on the **Mass Ingestion** page in Monitor. To download the error log, on the **Actions** menu for the job, click **Error Log**. After you resolve the issue, deploy the task again from the application ingestion and replication task wizard or from the **Actions** menu for the job.

## Step 7: Run the application ingestion job

You can run an application ingestion and replication job from the **My Jobs** page in the Data Ingestion and Replication service or from the **All Jobs** tab on the **Mass Ingestion** page in Monitor.

For initial load jobs, you can specify a schedule for running the job when you configure the corresponding application ingestion and replication task.

▶ On the **Actions** menu for the job that you want to run, click **Run**.

  A subtask is started for each source object.

  **Notes:**

  - If an initial load job fails to load data to a target table, the application ingestion and replication job retries the subtask for the object up to three times. The minimum interval between the retries is 60 seconds. If all the initial load retries fail, Application Ingestion and Replication excludes the object from replication.

- If an initial load job detects inconsistencies between field definitions in the source and target objects, the job drops the target table and then re-creates it to be consistent with the source object before loading the source data to the target.
- Initial load jobs may take a long time to complete the ingestion if the source objects contain many records.

# Author

**Informatica Documentation Team**